

A Medium-Term Integrated Risk Management Model for a Hydrothermal Generation Company

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Abstract—This paper presents a methodology to manage the market risk faced by a hydrothermal generation company in the medium-term (one year). This risk is due to uncertainty in fuel prices, power demand, water inflows, and electricity prices. The proposed methodology includes three steps: the generation of scenarios for these random parameters, the approximation of these scenarios by a multivariate scenario tree, and the optimization of the company's operational and financial hedging decisions under a stochastic programming framework. The optimization model permits the representation of a diversified generation portfolio and measures risk exposure by means of conditional value-at-risk. A realistic numerical example is solved to illustrate the possibilities of our approach.

Index Terms—Conditional value-at-risk, generation operation planning, integrated risk management, risk analysis, stochastic optimization.

I. INTRODUCTION

DURING the last decade, the electric power industry has undergone an intense process of reforms. In the new deregulated context, power generation companies are exposed to the risks associated with uncertain factors inherent to their business. These risks arise in all the activities that constitute the generation business from both an operational and a financial perspective [1].

In the short term (one day to one month), a generation company faces operational risks associated with uncertain factors such as the unexpected failure of a generation unit or the fluctuation of demand due to, for example, unforeseen temperature changes. Additionally, it is exposed to financial risks associated with the volatility of electricity prices [2].

In the medium term (several months to a few years), operational risks can be identified in the procurement of the minimum required amount of fuel or the availability of natural resources due to uncertain water inflows. From the financial perspective, the variety of risks is much wider than in the short term. The company is exposed to market risk associated with the uncertain evolution of forward electricity and fuel prices. Credit risk is due to the possibility of counter-party default. Currency risk is suffered by companies engaged in transactions with different

currencies, so that the value of their portfolios is affected by changes in exchange rates. Liquidity risk is related to changes in the bid-ask spread on futures transactions.

In the long term (more than three years), operational risks are related, for example, to the construction of generation plants. The role of financial risks is even more important than in the medium term. The impact of changes in interest rates is a good example.

This paper focuses on the medium-term problem faced by a generation company when planning the operation of its units in the presence of uncertainty. This problem should include not only pure operational decisions, but also risk management decisions. Risk management is the process of analyzing exposure to risk and determining how to best handle such exposure. There are financial instruments specifically designed to manage risk such as futures contracts, options, etc. Additionally, production resources can be used to manage risk to a certain extent. This combined perspective is what some authors have named *integrated risk management* [3]. The combination of these two aspects of the generation business in a decision-support model requires the explicit representation of the operation of the generation units, as well as measuring the risk exposure of the company due to the mentioned sources of uncertainty.

The integrated risk management problem has already been considered by different authors (see [4]–[6]). References [4] and [5] represent the generation system with special emphasis on hydro production and uncertainty of water inflows, while treating thermal resources, subject to fuel price uncertainty, in less detail or not considering them at all. In contrast, [6] mainly focuses on thermal generation, particularly CCGTs. However, all these authors neglect the strategic interaction of generation companies in the wholesale electricity market. As a consequence, they isolate the company from its competitors and assume that the uncertain spot price of electricity is independent of the company's operation strategy and the risk management decisions. In this paper, we propose a methodology that explicitly considers this interaction by means of an equilibrium model.

With respect to the risk measuring methodology used to guide the risk management process, other authors consider two different approaches. In [4] and [5], each scenario in which the company's profit is lower than a certain objective value is penalized. This has the advantage of not requiring a constraint that couples the different scenarios (when a scenario exceeds the loss limit it is penalized irrespective of the situation of the rest of scenarios). A disadvantage of this method is that it does not impose an explicit limit for the losses that the company may

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face. In [6] conditional value-at-risk (*CVaR*) is used as risk measure. This establishes a hard limit on the expected value of losses for the subset of scenarios in which the shortfall exceeds the value-at-risk (*VaR*) at a given confidence level [7]. For a comprehensive theoretical and computational comparison and classification of some risk measures see [8]. In this paper, we have chosen *CVaR* as risk measure.

The mathematical formulation of the medium-term integrated risk management problem of a generation company leads to a multistage stochastic program that can be solved using different techniques. In [5], the stochastic dual dynamic programming (SDDP) approach proposed in [9] is used in order to deal with such large-scale problems. In addition, [10] formulates a general multistage stochastic programming problem with mixed integer variables, using *CVaR* as risk measure, highlighting a two-stage procedure to solve the problem. In contrast, in [4] and [6], the equivalent deterministic problem is solved with a commercial optimizer.

This paper represents a step forward with respect to previous approaches in the field of managing the medium-term risk faced by a hydrothermal generation company. We propose a methodology with the following salient features. Firstly, we consider uncertainty in a variety of factors, such as water inflows, demand, electricity prices, fuel prices, and exchange rates. Secondly, we generate scenarios for the wholesale electricity market taking into account the effects of the strategic interaction between generation companies by means of an equilibrium model. Thirdly, we represent the uncertainty faced by the generation company in the form of a scenario tree constructed with state-of-the-art techniques. Fourthly, we formulate the integrated risk-management problem as a multistage stochastic linear programming model that we solve via its equivalent deterministic problem with a commercial optimizer. Although the model permits a risk penalization approach, we have specifically designed it to use *CVaR* as risk measure due to its better risk and mathematical properties.

The paper is organized as follows. Section II presents a general description of the considered problem, whereas Section III explains in detail the methodology we have used to tackle this problem. Section IV presents the mathematical formulation of the integrated risk management model we have developed. Section V illustrates the performance of this model with a realistic numerical example. Finally, Section VI summarizes the main conclusions of our research.

II. PROBLEM DESCRIPTION

In this section, we describe the medium-term market risk management problem of a hydrothermal generation company.

A. Medium-Term Hydrothermal Coordination

In a regulated environment, the medium-term hydrothermal coordination problem consists of satisfying the demand for power of a certain region over a time horizon ranging from several months to a few years at the minimum generation cost. Unless strictly necessary, network constraints are neglected. In general, hydro reserves are scarce and reservoirs have lower

and upper limits. This complicates the decision of when is best to use hydro resources to substitute thermal generation. The explicit consideration of fuel stocks increases the complexity of the problem. Researchers have developed complicated models to address this problem and powerful optimization techniques to obtain numerical results [11]–[14].

In recent years, the hydrothermal coordination problem in a competitive context has been addressed as an equilibrium problem. In [15] it is shown how generation companies should manage their hydro resources so that their water value equals their thermal marginal cost and their marginal revenue in the wholesale electricity market. In [16], a dual dynamic programming approach is proposed in which the problem of each stage is modeled as a Cournot equilibrium. In [17] and [18], a realistic hydrothermal coordination problem, including an explicit representation of the wholesale market equilibrium, is formulated and solved as a complementarity problem.

In this paper, we address the medium-term integrated risk-management problem of a hydrothermal generation company. The combination of a risk-management perspective with the classical hydrothermal coordination approach poses modeling and computational difficulties. For the sake of clarity, we have intentionally avoided the representation of certain technical aspects of the operation of thermal units or hydro plants. For example, we are using an equivalent model that comprises all the hydro generation units located in the same basin, which implies that we are not taking into account head effects or complications such as the presence of hydro units owned by other companies in the same basin. These simplifications indicate that our methodology is not specifically oriented to derive a detailed medium-term generation schedule for the company of interest, but rather to suggest medium-term guidelines both from a risk-management and an operational point of view, in accordance with the integrated perspective that we propose.

B. Sources of Uncertainty

As mentioned, the medium-term hydrothermal coordination problem of a generation company is subject to several sources of uncertainty. This suggests adopting a stochastic programming framework.

There are sources of uncertainty that are present in the medium term irrespective of the regulatory framework. These are water inflows, the evolution of the demand for power and of fuel prices, the availability of generation units, and the variability of exchange rates. Some of these uncertain factors might not imply risks for generation companies in a regulated context. However, this has radically changed with deregulation. In this paper, we consider these aspects using two different approaches. On the one hand, we incorporate the stochasticity of water inflows, demand and fuel prices into the construction of the scenario tree. Although we do not include them in the model formulation, we treat currency exchange rates essentially in the same way. On the other hand, we reduce the net maximum power output of each thermal unit in each month to its historic average available capacity. This yields more realistic results in terms of spot prices for on-peak hours and energy produced by peaking units.

With the advent of competition, new sources of uncertainty arise due to the unpredictable behavior of agents when participating in the different market mechanisms. This leads to uncertain wholesale electricity prices and uncertain sales of different products (energy, reserve, capacity). In this paper we take into account the uncertainty of the amount of energy that the generation company is able to sell at each load level, as well as of the price at which this energy is sold in the construction of the scenario tree.

C. Integrated Risk Management

A generation company typically prepares annual budgets which include an estimate of the profits obtained during the year. In addition, the company must be able to evaluate whether the probability of facing a low-profit outcome is higher than desired. This requires analyzing the operating profit of the generation company under a relevant number of uncertainty realizations.

This simulation approach would yield a probability distribution function for the annual profit of the generation company. Given this probability distribution, one could consider the possibility of finding a hedging strategy to improve the worst scenarios.

As mentioned above, there are two mechanisms that a generation company can use in order to manage market risk. The first one is to participate in futures markets for the different underlying assets (electricity, fuel, currency). The second one is to change its operational decisions, which implies that the probability function of the operating profit should be constructed at the same time as the hedging strategy is decided. This is what we understand by integrated risk management.

Current planning practice in many generation companies is far from this integrated perspective. In many cases, operation planning decisions are taken without considering the market risk exposure of the generation company. In this context, a methodology such as the one we propose may serve as a first step toward the objective of combining both points of view.

III. PROPOSED METHODOLOGY

This section presents the methodology we propose in order to address the integrated medium-term risk management problem of a hydrothermal generation company (Fig. 1).

A. Scenario Generation

Scenarios are possible realizations of the stochastic variables up to the end of the time horizon. As usual in stochastic programming, we discretize continuous time into periods of time. In this paper, we consider a one-year time horizon and divide it into different time periods depending on the random variable under consideration. In particular, we distinguish two types of random variables: electric market variables and exogenous variables.

Random electric market variables include power demand, electricity prices and power produced by the company. We discretize electric market variables into load levels so that

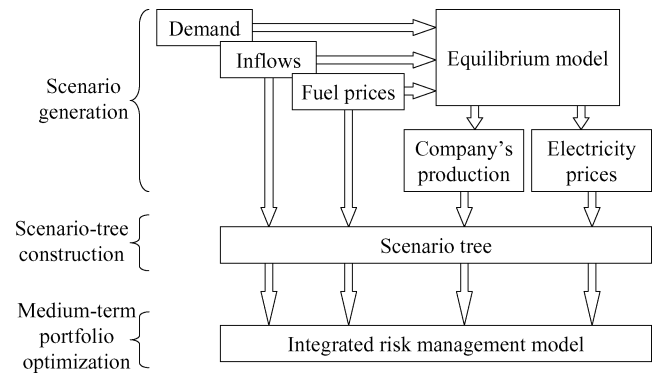


Fig. 1. Proposed approach.

each month includes several of these load levels (Fig. 2). This permits representing the typical short-term volatility of power prices, although we assume that uncertainty unfolds in monthly periods. We discretize the rest of random variables (fuel prices, water inflows, or exchange rates) in monthly periods.

Generating scenarios in this context requires a deep knowledge of the underlying probability distribution functions including correlations between the different random variables. In some cases, two variables may be correlated because they share a common explanatory variable (e.g., water inflows and demand both depend on weather). In other cases correlation is due to a cause-effect relationship (e.g., fuel prices, water inflows and demand have a clear impact on electricity prices).

It is beyond the scope of this paper to provide a full description of the methodology followed to generate the scenarios for power demand, water inflows or fuel prices (see [19]).

In contrast, one of the main contributions of this paper is to propose the use of a fundamental model (in particular, an equilibrium model [17]) rather than an econometric model to derive the scenarios for electricity prices and the power produced by the company. This contribution can be justified from a number of points of view.

- It is evident that both approaches can model the influence of fuel prices, water inflows and power demand on electricity spot prices. An econometric model can incorporate this influence through some type of correlation based on historical data, whereas a fundamental model uses fuel prices, water inflows and power demand to determine the power output of the different generation companies and, as a byproduct, the spot price of electricity.
- In principle, an econometric model cannot foresee the correlation between variables for a situation that is structurally different from one observed in the past (e.g., if a new generation plant begins to operate at a certain moment of the year or if a merger between two generation companies takes place). An equilibrium model provides an answer to this question.
- Deciding the power output of the generation company in each load level using as input data the electricity spot prices given by an econometric model is not an easy task. The spot price of electricity and the power output of the different generation companies are the result of their interaction in the wholesale market. Hence, it seems more

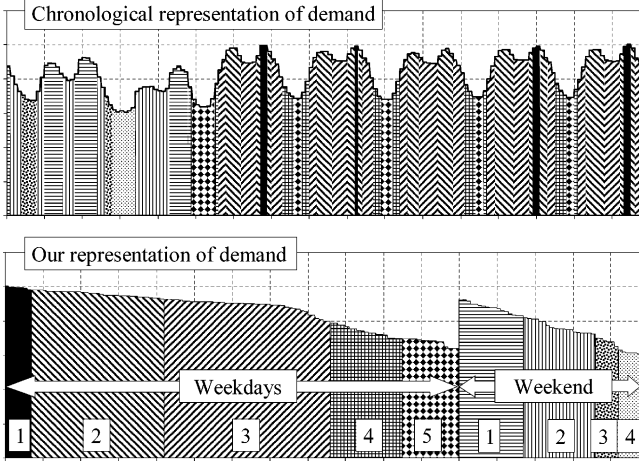


Fig. 2. Discretization of one week into load levels.

reasonable to simultaneously obtain both types of variables (prices and power outputs). We do so by means of the equilibrium model.

The use of equilibrium models to represent competition between generation companies in wholesale electricity markets is nowadays a mature line of research [20]. Among a variety of modeling approaches, Cournot model of equilibrium has received much attention. In this framework each generation company e is assumed to decide the production level q_e that maximizes its profit given the production level of the rest of companies. The price of electricity is expressed as a decreasing function of the total amount of power generated by the companies. This function (the inverse demand function) establishes a link between the decisions taken by the different companies thus representing their interaction in the wholesale electricity market (Fig. 3).

When applying Cournot equilibrium model to the case of power generation, researchers have included a more or less detailed representation of the generation units so that a byproduct is a production profile for each of the companies. This is one of the outputs of our equilibrium model that we are interested in (Fig. 1).

With respect to the other output, pure Cournot equilibrium models tend to provide higher prices than those observed in reality. Researchers have addressed this problem by introducing the concept of conjectured variations which aims to represent the fact that rivals do react to high prices by producing more [21], [22].

Nevertheless, the manner in which we use the equilibrium model to derive scenarios for the spot price of electricity and for the company's power output implies some important simplifying assumptions.

- We are obtaining each annual scenario of the spot price of electricity and of the company's output by formulating and solving the deterministic equilibrium problem corresponding to one annual scenario of fuel prices, hydro inflows and demand for power. This is obviously a simplification because, in this setting, generation companies owning hydro reservoirs can decide in which month it is

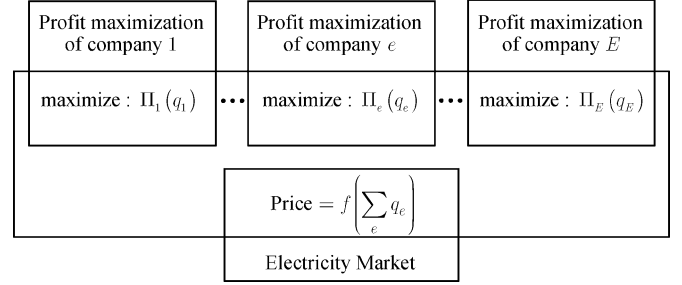


Fig. 3. Equilibrium computing of a power generation market.

more profitable to use its hydro reserves with perfect information about the future.

- We are assuming that the company's power output for each load level will not change in the risk management stage of our methodology. In other words, in the risk management stage we consider as input data the company's output given by the equilibrium model for each load level and each scenario. Hence, the risk management model decides the generation units that should produce this power (hydrothermal coordination), e.g., shifting hydro production between months to reduce the level of risk exposure (integrated risk management). The underlying assumption is that the impact of the companies' hedging strategies on the market equilibrium is small.

B. Multivariate Scenario-Tree Construction

Once the generated scenarios represent accurately enough the space of probability for the medium-term integrated risk management problem, we suggest the construction of a multivariate scenario tree. The random variables considered for this scenario tree are the company's power production, the spot price of electricity, the spot price of the different fuels, and water inflows. A scenario tree provides a representation of uncertainty that permits the application of stochastic programming techniques [23].

1) *Scenario-Tree Construction Techniques:* The problem of scenario-tree construction has been addressed by many authors. [24] propose a heuristic moment-matching method that is specifically used to find the discrete tree-structured distribution whose statistical parameters *best approximate* those of the underlying distribution. [25] and [26] extend the concept of clustering techniques to obtain a collection of representative scenarios that are forced to share part of their nodes up to predetermined ramification nodes. The technique we have used to construct scenario trees is similar to the one described in [26].

2) *Treating Arbitrage Opportunities:* The construction of a scenario tree with the purpose of evaluating the possibility of entering into forward contracts to hedge against the volatility of prices in the spot market requires some discussion on the relationship between forward and spot prices.

If we denote random variables with bold characters, let $\mathbf{S}_t$ be the spot price for each time period t , and $\mathbf{F}_{l,t}$ the forward price in period l for delivery in period t (Fig. 4). In order to evaluate a hedging operation in the first node of the scenario tree with delivery in period t , the company must compare the spot price

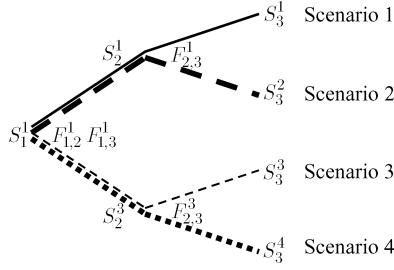


Fig. 4. Two-stage four-scenario tree.

expected for period t , $\mathbb{E}(S_t)$, with the forward price in the first period for delivery in period t , $F_{1,t}$. If $F_{1,t} > \mathbb{E}(S_t)$ then there is an arbitrage opportunity: The company can increase its expected profit by making a forward sale of $f_{1,t}^s$. The only costs of this operation are the transaction costs $c_{1,t}^f(f_{1,t}^s)$. On the other hand, if $F_{1,t} < \mathbb{E}(S_t)$ the company can increase its expected profit by buying $f_{1,t}^b$ forward contracts. In both cases the company increases its expected profit without increasing its risk exposure.

In this paper, we consider a nonarbitrage approach in order to avoid arbitrage opportunities. This implies that forward prices must be equal to the corresponding expected spot prices. A difficulty arises in the first node of the tree for which forward prices are already known. It is likely that the expected spot price for a certain time period t seen from this first node, $\mathbb{E}(S_t)$, is not equal to the current forward price for delivery in t , $F_{1,t}$. This requires adjusting the scenario tree, which can be done by modifying either transition probabilities or spot-price scenarios [26]. By doing so, the expected spot price for a certain time period is forced to be equal to the current forward price for that period.

An alternative would be to admit that there can be arbitrage opportunities due to insufficient liquidity in the forward market or to information asymmetry between the different agents. Under the assumption of limited liquidity, transaction costs in the forward market should be high, in order to limit the volume of contracts that is profitable to trade.

C. Medium-Term Portfolio Optimization

Given the scenario tree, the medium-term integrated risk-management problem can be formulated and solved as a multistage stochastic programming problem, including both operational and risk management decisions.

1) *Representation of the Generation Portfolio*: The representation of the medium-term operation of a generation system has been extensively treated both in a competitive and a noncompetitive framework. However, when addressing the risk management problem of a generation company, authors have focused on a particular generation technology, either thermal [2], [6] or hydro [4], [5].

In our model, we represent in detail the operation of the generation units of a company with a diversified portfolio, taking as input data both the power that the company produces in each load level and each scenario, which is a step forward with respect to previous works. The power output of each generation unit in each scenario and the expected profit obtained by the company in the electricity market are model outputs.

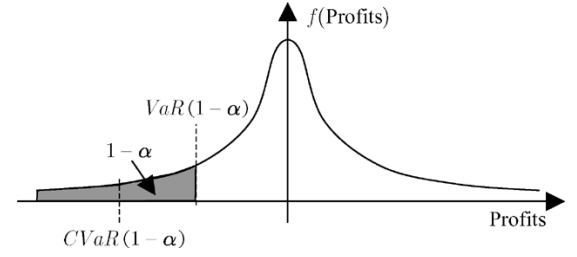


Fig. 5. CVaR concept.

2) *Representation of the Financial Portfolio*: Our representation of the medium-term financial portfolio of a generation company takes as input data the forward and spot prices of different commodities (electricity, fuel, etc.) The possibility of entering on forward contracts for commodities other than electricity (fuel, currencies, etc.) is also an original feature of our model.

We manage risk in our model by imposing a lower limit on $CVaR$. For a given time horizon T and confidence level α , $CVaR(1 - \alpha)$ is the conditional expectation of the profit above $VaR(1 - \alpha)$. As an alternative to VaR , $CVaR$ is attractive since it is a coherent risk measure [27]. The concept of $CVaR$ is illustrated in Fig. 5.

$CVaR$ as a risk measure has interesting properties such as translation invariance, subadditivity, positive homogeneity, monotonicity, relevance, consistency with first- and second-order stochastic dominance, and convexity. Second-order stochastic dominance represents risk aversion for a rational agent, whereas convexity is a remarkable mathematical programming property. Moreover, $CVaR$ (for discrete random variables) can be modeled by means of linear expressions. The several advantages of $CVaR$ in the optimization of a power portfolio have already been described in [1].

IV. MODEL FORMULATION

This section presents the formulation of the integrated risk-management problem. Parameters and variables are denoted with upper and lower case symbols, respectively. Stochastic parameters and variables are represented in bold symbols.

A. Representing Uncertainty

As we have explained in Section III, continuous time has been discretized into periods, t to T , and the uncertainty of parameters and variables has been represented by means of a multivariate scenario tree. This approach assumes that the decision process is nonanticipative.

In order to represent a scenario tree in a mathematical programming problem, there are several possibilities. One choice is to formulate a deterministic problem for each scenario and then incorporate nonanticipativity constraints, which forces the decisions taken in each node shared by two or more scenarios to be equal. An alternative to this scenario-based formulation is to adopt a node-based formulation specifying the relationship that exists between each node and its ancestor. For simplicity, the formulation presented in this paper does not explicitly follow

either of these two approaches, although it does highlight stochastic parameters and variables with bold characters.

B. Objective Function

As mentioned in Section III, the integrated risk management problem maximizes the expected value of the company's profits, Π , subject to certain risk exposure limitations

$$\max \quad \mathbb{E}(\Pi). \quad (1)$$

The company's annual profit is calculated as the sum of the company's profit in each time period t , Π_t

$$\Pi = \sum_t \Pi_t. \quad (2)$$

Π_t is obtained as the sum of the profits due to the operation of the company's generation units, Π_t^o , and the profits due to the financial contracts signed by the company in each period $l < t$ for delivery in period t , $\Pi_{l,t}^f$

$$\Pi_t = \Pi_t^o + \sum_{l < t} \Pi_{l,t}^f. \quad (3)$$

1) *Operation Profits:* Π_t^o is calculated as the difference between operation revenues, $\mathbf{r}_t^o$, and operation costs, $\mathbf{c}_t^o$

$$\Pi_t^o = \mathbf{r}_t^o - \mathbf{c}_t^o. \quad (4)$$

Operation revenues are given by

$$\mathbf{r}_t^o = \sum_n D_{t,n} \cdot \mathbf{S}_{t,n}^e \cdot \left(\sum_i \mathbf{q}_{t,n}^i + \sum_j \mathbf{q}_{t,n}^j \right) \quad (5)$$

where $D_{t,n}$ is the duration of each load level n , in hours; $\mathbf{S}_{t,n}^e$ is the spot price of electricity, in Euro/MWh; $\mathbf{q}_{t,n}^i$ represents the power output of each thermal unit i , in MW; and $\mathbf{q}_{t,n}^j$ the power output of each hydro unit j , in MW.

Operation costs are calculated as

$$\mathbf{c}_t^o = \sum_n D_{t,n} \cdot \sum_i \mathbf{q}_{t,n}^i \cdot \left(O^i + \mathbf{S}_t^{c(i)} \cdot V^i \right) \quad (6)$$

where O^i are unit i 's variable O&M costs, in Euro/MWh; V^i is unit i 's average fuel consumption, in MWh (HHV)/MWh; and $\mathbf{S}_t^{c(i)}$ is the spot price of unit i 's fuel, c , in each time period t , in Euro/MWh (HHV).

2) *Financial Profits:* $\Pi_{l,t}^f$ quantifies the company's profits due to its financial contracts signed in period l for delivery in period t over underlying assets such as electricity and different fuels

$$\Pi_{l,t}^f = \Pi_{l,t}^{f,e} + \sum_c \Pi_{l,t}^{f,c}. \quad (7)$$

In both types of commodities, profits due to financial contracts are calculated as the difference between revenues and costs

$$\Pi_{l,t}^{f,e} = \mathbf{r}_{l,t}^{f,e} - \mathbf{c}_{l,t}^{f,e} \quad (8)$$

$$\Pi_{l,t}^{f,c} = \mathbf{r}_{l,t}^{f,c} - \mathbf{c}_{l,t}^{f,c}. \quad (9)$$

In order to calculate the company's revenues due to the settlement of forward electricity contracts, we assume that the company has a net selling position. In each time period l , the company has the possibility of making a forward sale (purchase) of power for delivery in period t and load level n . We denote the amount of power implied in such a forward sale (purchase) by $\mathbf{f}_{l,t,n}^{e,s}$ ($\mathbf{f}_{l,t,n}^{e,b}$), in MW. Such an operation will yield a revenue that is proportional to the difference between the price of the forward contract $\mathbf{F}_{l,t,n}^e$ and the spot price of electricity in period t , $\mathbf{S}_{t,n}^e$. Hence, the company's revenues due to the settlement of its financial contracts signed in period l for delivery in period t are calculated as follows:

$$\mathbf{r}_{l,t}^{f,e} = \sum_n D_{t,n} \cdot \left\{ (\mathbf{F}_{l,t,n}^e - \mathbf{S}_{t,n}^e) \cdot (\mathbf{f}_{l,t,n}^{e,s} - \mathbf{f}_{l,t,n}^{e,b}) \right\}. \quad (10)$$

In contrast, when calculating the company's revenues due to the settlement of forward fuel contracts we assume that the company has a net buying position

$$\mathbf{r}_{l,t}^{f,c} = D_t \cdot \left\{ (\mathbf{S}_t^c - \mathbf{F}_{l,t}^c) \cdot (\mathbf{f}_{l,t}^{c,b} - \mathbf{f}_{l,t}^{c,s}) \right\} \quad (11)$$

where D_t is the duration of period t , in hours; $\mathbf{S}_t^c$ is the spot price of fuel c in period t , in Euro/MWh; $\mathbf{F}_{l,t}^c$ is the forward price of fuel c in period l for delivery in period t , in Euro/MWh; $\mathbf{f}_{l,t}^{c,b}$ ($\mathbf{f}_{l,t}^{c,s}$) represents the purchases (sales) of fuel c forward, in megawatts, in period l for delivery in period t .

In both kinds of underlying assets, transaction costs include not only the bid-ask spread, but also costs associated to liquidity risk. We model transaction costs as a convex piecewise linear function of the traded volume of forward contracts. In the case of forward electricity contracts

$$\mathbf{c}_{l,t}^{f,e} \geq \sum_n D_{t,n} \cdot \left\{ C_m^{f,e} \cdot (\mathbf{f}_{l,t,n}^{e,s} + \mathbf{f}_{l,t,n}^{e,b}) + C_m^e \right\} \quad (12)$$

where m are the different sections of the piecewise linear function; $C_m^{f,e}$ is the slope of each section m , in Euro/MWh; and C_m^e is the y -intercept, in Euro/h.

A similar expression is used for fuel contracts

$$\mathbf{c}_{l,t}^{f,c} \geq D_t \cdot \left\{ C_m^{f,c} \cdot (\mathbf{f}_{l,t}^{c,s} + \mathbf{f}_{l,t}^{c,b}) + C_m^c \right\}. \quad (13)$$

C. Generation Representation

1) *Company's Power Output:* As mentioned, the company's power output in each load level, $\mathbf{Q}_{t,n}$, is considered as input data

for the integrated risk management problem. This power output is provided by the company's thermal and hydro units

$$\sum_i q_{t,n}^i + \sum_j q_{t,n}^j = Q_{t,n} \quad \forall t, n. \quad (14)$$

2) *Water Reservoir Management*: A generation company owning hydro reservoirs has the possibility of managing hydro reserves to partially hedge against low-profit scenarios. For example, hydro reserves can be saved to protect against particularly dry scenarios, although this may increase the probability of spilling water if a particularly wet scenario finally occurs. Hence, we consider the amount of water in each reservoir j at the end of each month t to be a variable, $w_{t,j}$, that depends on hydro production, $q_{t,n}^j$, and inflows, $I_{t,j}$, during that month as well as of the level of reserves at the end of the previous month:

$$\sum_n (D_{t,n} \cdot q_{t,n}^j) \leq I_t^j + w_{t-1}^j - w_t^j \quad \forall t, j. \quad (15)$$

3) *Take-or-Pay Contracts*: We assume the company has entered into one natural gas take-or-pay contract for each gas unit g in each period t

$$\underline{T}_t \leq \sum_{n,g} D_{t,n} \cdot q_{t,n}^g \leq \bar{T}_t \quad \forall t \quad (16)$$

where $\underline{T}_t$ and $\bar{T}_t$ are the minimum and maximum quantities of natural gas implied by the take-or-pay contract for period t , in MWh.

D. Financial Portfolio Representation

As mentioned, in order to avoid arbitrage opportunities, we force forward prices in periods different from the first one to be equal to the corresponding expected spot prices

$$F_{l,t,n}^c = \mathbb{E}(S_{t,n}^c) \quad \forall l > 1, t > l, n \quad (17)$$

$$F_{l,t}^c = \mathbb{E}(S_t^c) \quad \forall l > 1, t > l, c. \quad (18)$$

E. Risk Modeling

As mentioned, we manage risk by means of a risk measure, $CVaR$, which can be modeled with linear expressions (see [6] and [7]).

Let Π be a random variable representing the profits of a certain portfolio. If F_Π is the probability distribution function of Π , the VaR for a confidence level α , $VaR_\Pi(1 - \alpha)$, is given by

$$VaR_\Pi(1 - \alpha) = \min\{u : F_\Pi(u) \geq 1 - \alpha\}. \quad (19)$$

The mean level of profits lower than $VaR_\Pi(1 - \alpha)$ is known as $CVaR_\Pi(1 - \alpha)$

$$CVaR_\Pi(1 - \alpha) = \mathbb{E}(\Pi \mid \Pi \leq VaR_\Pi(1 - \alpha)). \quad (20)$$

In order to formulate $CVaR$ by means of linear expressions, we define a new random positive variable z , which measures negative profit deviations from $VaR_\Pi(1 - \alpha)$

$$z \geq VaR_\Pi(1 - \alpha) - \Pi. \quad (21)$$

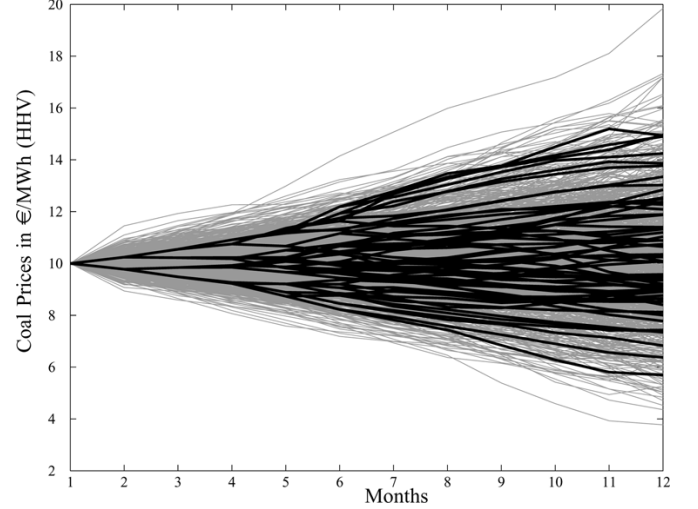


Fig. 6. Coal prices.

Using this negative variable, $CVaR$ can be calculated as

$$CVaR_\Pi(1 - \alpha) = VaR_\Pi(1 - \alpha) - \frac{1}{1 - \alpha} \cdot \mathbb{E}(z) \quad (22)$$

where $CVaR_\Pi(1 - \alpha)$ is the expected profit, in Euro, of those scenarios with profits lower than $VaR_\Pi(1 - \alpha)$.

V. CASE STUDY

The model formulated in this paper has been developed and implemented in GAMS [28]. In this section, we present the results obtained for a realistic numerical example in which we address the risk management problem of a fictitious Spanish hydrothermal generation company during year 2004. We have created this fictitious company by aggregating generation units belonging to different real companies.

A. Scenario Generation

The first step of our methodology is to generate scenarios for the random variables. We have generated 1000 possible realizations for fuel prices, water inflows, and power demand. We have considered the correlation between the different variables (e.g., between natural gas prices and oil prices) and the correlation between time periods (e.g., between hydro inflows in two consecutive months). The realizations of the different random variables (except power demand) are represented in Figs. 6–9, in light gray.

Considering these random variables as input data, we have used an equilibrium model to obtain, for each scenario, the company's power production and the spot price of electricity, for each load level in each time period. The scenarios obtained for both random variables are represented in Figs. 10 and 11, respectively, in light gray. As can be seen, our discretization of time into load levels (Fig. 2) leads to price and production series with two spikes in each month, corresponding to the on-peak load level of weekdays and of weekends, respectively.

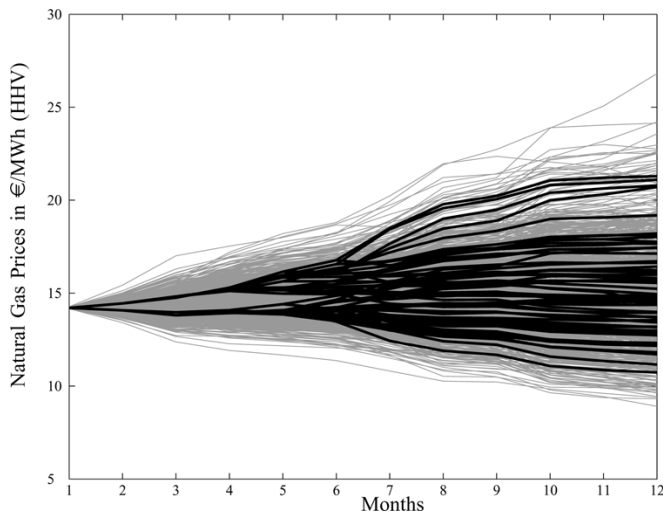


Fig. 7. Natural gas prices.

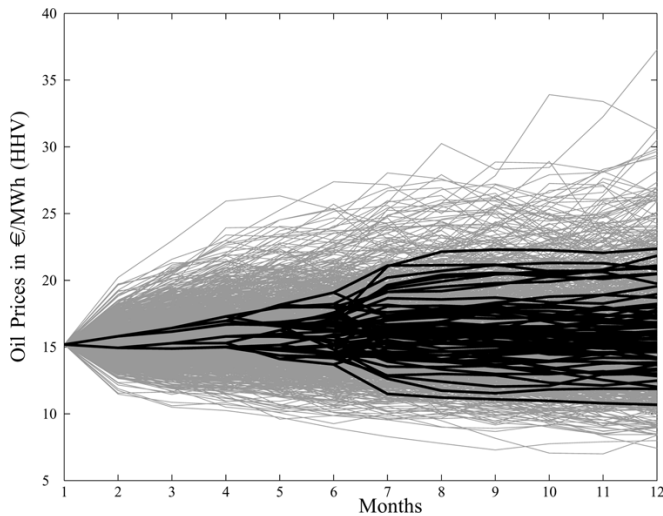


Fig. 8. Oil prices.

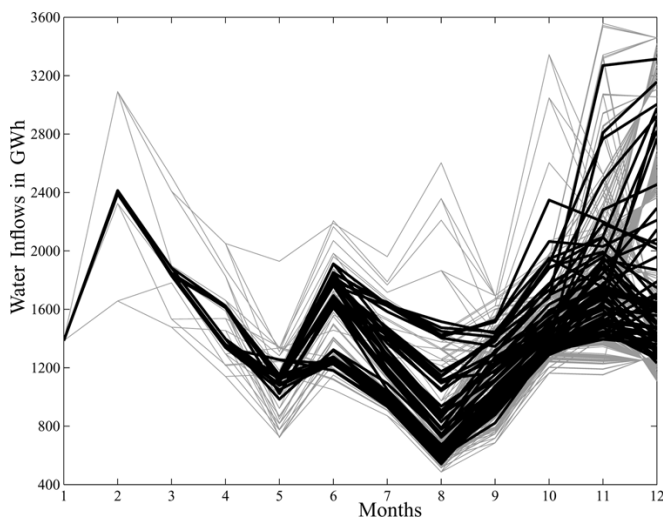


Fig. 9. Water inflows.

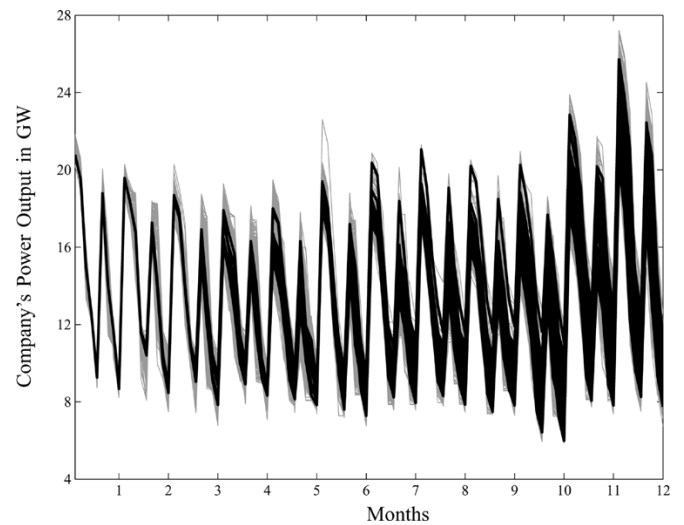


Fig. 10. Company's power output.

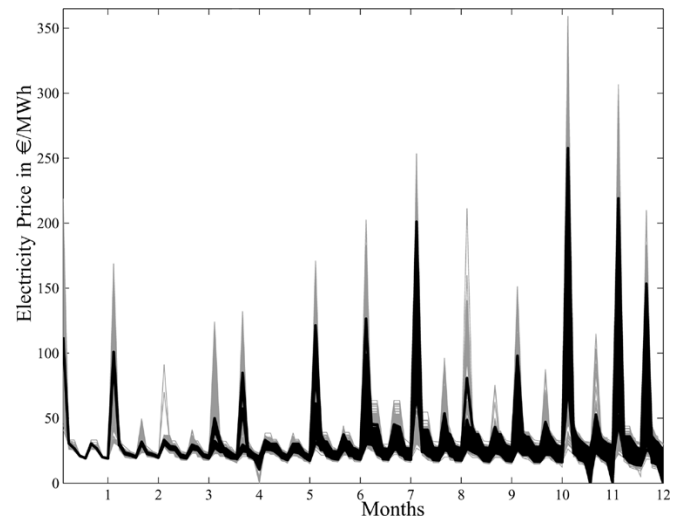


Fig. 11. Electricity price.

B. Scenario-Tree Construction

Once these 1000 multivariate scenarios have been generated, we have clustered them into the multivariate 64-scenario tree of Fig. 12 using techniques similar to those described in [26]. The resulting scenario tree is represented in Figs. 6–11. It is important to notice that each figure represents the same scenario tree under different perspectives. It is clear that the scenario tree provides an approximation that seems better for some variables (e.g., coal prices) than for others (e.g., oil prices), although it is the best that we have obtained for the set of variables with this tree structure.

C. Portfolio Optimization

The last step is to perform the optimization of the portfolio of the generation company based on the resulting scenario tree. This results in a problem with 225 000 variables and 100 000 equations that we have solved in a Pentium IV, 1800-MHz (512 MB RAM) in 65 s.

Fig. 13 shows the efficient frontier obtained for this problem with different *CVaR* limits. This figure provides the maximum

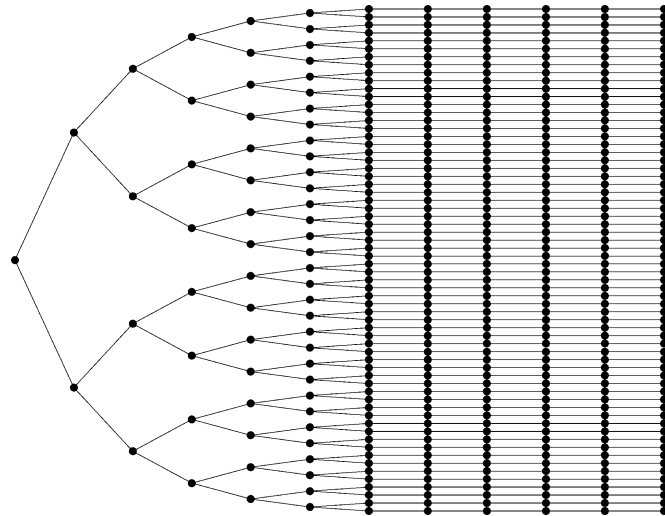


Fig. 12. Proposed scenario-tree structure.

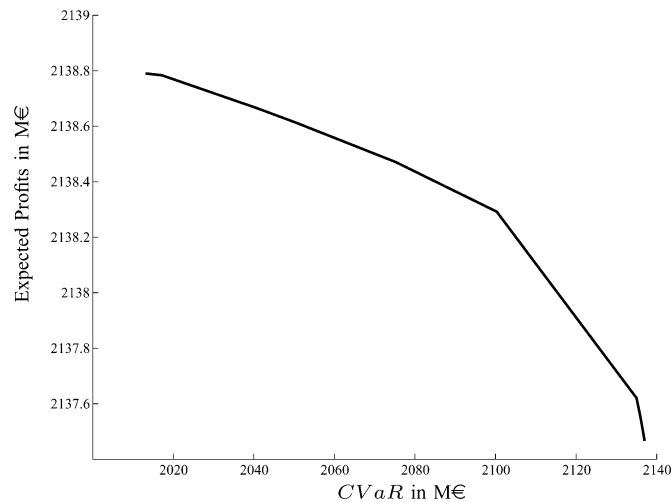


Fig. 13. Efficient frontier.

achievable expected profit for each level of risk ($CVaR$) or, alternatively, the minimum achievable level of risk ($CVaR$) for each expected profit. As can be seen, an increase of $CVaR$ of about 6% reduces the expected profit in 0.07%.

In Fig. 14, we represent the probability distribution functions of the profits that the company obtains for each level of risk ($CVaR$) imposed. As we increase the value of $CVaR$, the probability distribution of the company's profits gets narrower.

D. From Theory to Practice

The line of research followed by this and other related papers is part of the cultural change that is taking place in generation companies throughout the world as a result of the process of liberalization of the power industry.

The potential use of this type of methodology depends directly on the existence of liquid futures markets where the agents can carry out hedging operations with a relatively low cost.

In some cases, such as the Scandinavian power market, an organized futures market has been in operation for several years

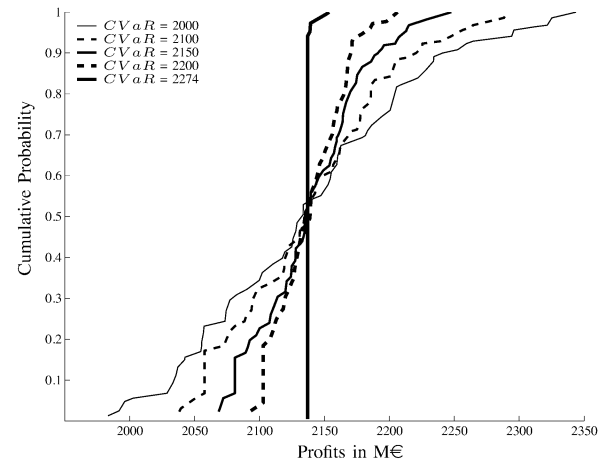


Fig. 14. Profits distribution function.

and has reached a satisfactory level of liquidity. This justifies the great interest shown both by academic and practitioners in this type of developments.

In the Spanish context, the future development of the Iberian Market of Electricity (MIBEL) will bring an Iberian Futures Market. The advent of such a marketplace will trigger the need for tools similar to the ones we describe. The development of a methodology to address the problems that will arise in this setting will surely be a valuable asset when the time comes.

VI. CONCLUSION

In this paper, we have presented a methodology to address the integrated risk management problem of a hydrothermal generation company. The methodology includes three main steps: the generation of scenarios for the relevant random variables, the approximation of these scenarios by means of a scenario tree, and the optimization of the company's decisions under a stochastic programming framework. Although this methodology may appear as the application of common sense, there are a number of relevant features that make our proposal original.

Firstly, we have adopted a combined approach to generate the scenarios for the random variables. For some of them (power demand, hydro inflows, fuel prices), we obtain scenarios using techniques that have already been developed by other authors. In contrast, we suggest using an equilibrium model that correctly reflects that the price of electricity can be different from the marginal cost of generation to obtain scenarios for electricity price and for the company's power production.

Secondly, our integrated risk management model permits the representation of a wide variety of generation technologies as well as the inclusion of contracts over underlying commodities such as electricity or fuels. It maximizes the company's expected profit subject to a risk exposure limitation. We evaluate this exposure with a coherent risk measure, $CVaR$.

Finally, we have tested our methodology with a realistic numerical example in the context of the Spanish electricity market. The results obtained in a reasonable computation time confirm the validity of our approach and offer a good perspective for further research.

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